

Causality - SoSe 2025

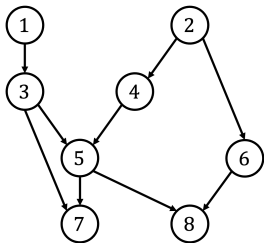
Causal Graphical Models

Alexander Marx

Last Time

In the last lecture, we discussed the following concepts:

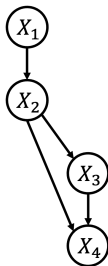
- Graph terminology
- Markov properties
- DAG models/Bayesian networks
- d-Separation



- What are the parents and the ancestors of node 5?
- What are all non-descendants of 4?
- Are 1 and 8 d-connected?
- Are 3 and 6 d-connected?
- Are 3 and 2 d-connected given 4?
- Does $3 \not\perp\!\!\!\perp_G 2 \mid \{4, 8\}$ hold?
- What is required that $X_5 \perp\!\!\!\perp X_2 \mid X_4$?

Definition (Global Markov Property) Given a DAG $\mathcal{G} = (\mathbf{V}, \mathcal{E})$. A distribution P over $\mathbf{X}$ with density p satisfies the **global Markov property** with respect to $\mathcal{G}$ if for all disjoint subsets $\mathbf{A}, \mathbf{B}, \mathbf{C} \subseteq \mathbf{V}$:

$$\mathbf{A} \perp\!\!\!\perp_{\mathcal{G}} \mathbf{B} \mid \mathbf{C} \implies \mathbf{X}_{\mathbf{A}} \perp\!\!\!\perp \mathbf{X}_{\mathbf{B}} \mid \mathbf{X}_{\mathbf{C}} .$$



Assume that P satisfies the global Markov property w.r.t. the graph on the left. Can we conclude that

$$X_1 \not\perp\!\!\!\perp X_4 \mid X_3 ?$$

Definition A **DAG model** or **Bayesian network** is a combination $(\mathcal{G}, P)$, where $\mathcal{G}$ is a DAG and P fulfills the Markov factorization property with respect to $\mathcal{G}$.

Outline

In today's lecture, we will learn about *causal* graphs. In particular, we discuss the following concepts:

- Structural Causal Models
- *do* Interventions
- Causal Bayesian Networks
- Average Treatment Effect

Structural Causal Models (SCMs)

Four Scenarios as Computer Programs (Reminder)

- If you get married, you are likely happier: $M = f(); H = g(M);$
- If you are happier, you are more likely to marry:
 $H = f(); M = g(H);$
- There is a common cause C that makes you more likely to be happy and to get married: $C = f(); M = g(C); H = h(C);$
- Being married and being happy has nothing to do with each other:
 $M = f(); H = g();$

Today, we will formalize this intuition!

Structural Causal Model

Definition A **structural causal model** $\mathcal{C} = (\mathbf{S}, P_{\mathbf{N}})$ is defined as a collection of assignments $\mathbf{S}$:

$$X_i := f_i(\mathbf{X}_{\text{Pa}_i}, N_i), \quad i = 1, \dots, d,$$

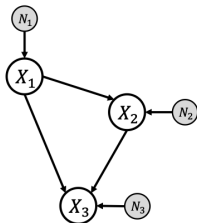
where all N_i are jointly independent so that $P_{\mathbf{N}}$ is a product distribution. The **graph** $\mathcal{G} = (\mathbf{V}, \mathcal{E})$ of an SCM is obtained by creating one vertex $j \in \mathbf{V}$ for each X_j and drawing directed edges $(i, j) \in \mathcal{E}$ from each parent $i \in \text{Pa}_j$ to j . We hence assume $\mathcal{G}$ to be **acyclic**.

Note that:

- An **SCM entails** a unique distribution $P_{\mathbf{X}}^{\mathcal{C}}$ over $\mathbf{X} = \{X_1, \dots, X_d\}$. Whenever clear from context, we write $P_{\mathbf{X}}$ instead of $P_{\mathbf{X}}^{\mathcal{C}}$.
- The nodes N_i are also called **noise sources**, **external**, or **exogenous variables**, and are often denoted as ϵ_i .

SCM (Example)

Directed Acyclic Graph (DAG)



Structural Causal Model (SCM)

$$X_1 := f_1(N_1)$$

$$X_2 := f_2(X_1, N_2)$$

$$X_3 := f_3(X_1, X_2, N_3)$$

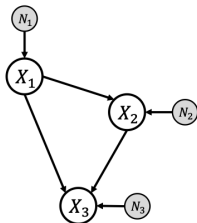
Each X_i is expressed as a **function of** its **parents** and an **external node**:

$$X_i := f_i(\mathbf{X}_{\text{Pa}_i}, N_i)$$

- N_1, N_2, N_3 (in gray) are **external** (not observed) random variables, or noise sources, which are **jointly independent**.
- X_1, X_2, X_3 are **observed RVs**, represented by nodes $\{1, 2, 3\}$.

SCM (Example)

Directed Acyclic Graph (DAG)



Structural Causal Model (SCM)

$$X_1 := f_1(N_1)$$

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$$X_3 := f_3(X_1, X_2, N_3)$$

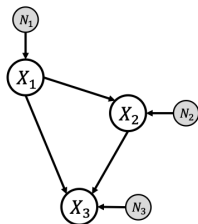
Each X_i is expressed as a **function** of its **parents** and an **external node**:

$$X_i := f_i(\mathbf{X}_{\text{Pa}_i}, N_i)$$

- The **SCMs encode** the **edges in the causal graph**, e.g., f_2 encodes the causal edge from X_1 to X_2 ($X_1 \rightarrow X_2$).
- Note that we use **assignments**, to indicating **asymmetry** introduced by the noise sources (different from laws in physics).

Structural Causal Model

Directed Acyclic Graph (DAG)



Structural Causal Model (SCM)

$$X_1 := f_1(N_1)$$

$$X_2 := f_2(X_1, N_2)$$

$$X_3 := f_3(X_1, X_2, N_3)$$

Each X_i is expressed as a **function** of its **parents** and an **external node**:

$$X_i := f_i(\mathbf{X}_{\text{Pa}_i}, N_i)$$

- The **functions** f_i can be very complex (e.g., one-way functions used in cryptography) or simple linear or non-linear additive functions
- Whenever clear from context, we **do not draw the external nodes** in the graph (i.e., we omit all gray nodes corresponding to N_i)

Linear Additive Noise Models

Definition A **linear additive noise model** is a SCM $\mathcal{C} = (\mathbf{S}, P_N)$ for which all assignments are **linear functions** and the **noise is additive**:

$$X_i := \mathbf{B}_{\text{Pa}_i} \mathbf{X}_{\text{Pa}_i} + N_i, \quad i = 1, \dots, d,$$

For example, let $N_i \sim \mathcal{N}(0, 1)$, and be pairwise independent. The linear SCM:

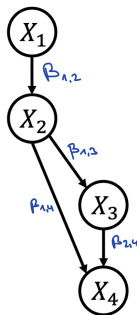
$$X_1 := \beta_{0,1} + N_1$$

$$X_2 := \beta_{0,2} + \beta_{1,2}X_1 + N_2$$

$$X_3 := \beta_{0,3} + \beta_{1,3}X_2 + N_3$$

$$X_4 := \beta_{0,4} + \beta_{1,4}X_2 + \beta_{2,4}X_3 + N_4$$

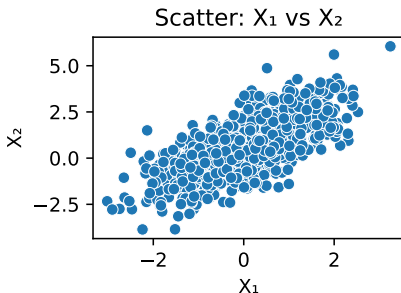
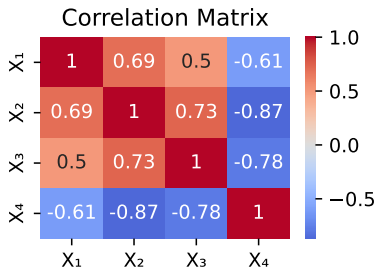
defines a distribution $P_{\mathbf{X}}$, and the DAG on the right.



Linear Additive Noise Models

```
n_samples = 1000
N = np.random.normal(0, 1, size=(n_samples, 4))

# Generate a linear SCM
X1 = N[:, 0]
X2 = 0.5 + X1 + N[:, 1]
X3 = 0.8 * X2 + N[:, 2]
X4 = -1 * X2 - 0.5 * X3 + N[:, 3]
```



Remark 6.6 (Structural Minimality, ECI Book) In an SCM, we assume that each structural assignment f_i **depends on all its inputs**.

Formally, whenever there is a $k \in \mathbf{V}$ and a function g such that

$$f_k(\mathbf{x}_{\text{Pa}_k}, n_k) = g(\mathbf{x}_{\text{Pa}_k^*}, n_k), \quad \forall \mathbf{x}_{\text{Pa}_k}, \forall n_k \text{ with } p(\mathbf{x}_{\text{Pa}_k}) > 0,$$

where $\text{Pa}_k^* \subsetneq \text{Pa}_k$, we choose the latter representation.

For example, we prefer $\mathcal{C}^1$ over $\mathcal{C}^0$ below:

$$(\mathcal{C}^0) \quad X := N_X, \quad Y := 0 \cdot X + N_Y$$

$$(\mathcal{C}^1) \quad X := N_X, \quad Y := N_Y$$

From now on, we **assume that structural minimality holds**.

SCMs entail the Markov Property

Proposition 6.31 (ECI Book) Assume that P_X is **induced by an SCM** with graph $\mathcal{G}$. Then, P_X is **markovian** with respect to $\mathcal{G}$.

Interventions

Interventions and Causal Effects

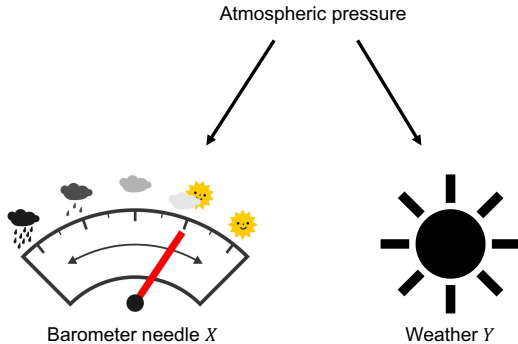


Figure 1: Credits Frederick Eberhardt

Interventions and Causal Effects

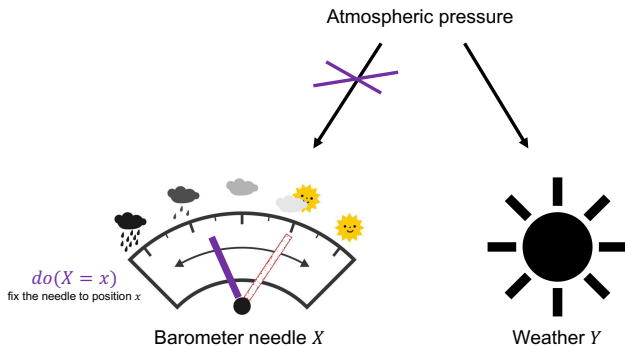


Figure 2: Credits Frederick Eberhardt

X has a **causal effect** on Y , if **manipulating** X **changes** Y .

Causal Effect and do-Operator

An **intervention** is a modification of an SCM, where we **replace one of the assignments**, i.e., we change the mechanism.

We distinguish:

- **Hard** interventions: $do(X = x)$, which replaces the SCM for X_i by $X := x$ (deletes all influences from parent nodes of X_i)

$$X := f_X(\mathbf{X}_{\text{Pa}_X}, N_X) \xrightarrow{do(X=x)} X := x$$

Causal Effect and do-Operator

An **intervention** is a modification of an SCM, where we **replace one of the assignments**, i.e., we change the mechanism.

- **Soft** interventions, which replace the structural assignment by a random variable, e.g.,

$$X := f_X(\mathbf{X}_{\text{Pa}_X}, N_X) \xrightarrow{\text{do}(X=N_X^*)} X := N_X^*$$

or change the whole mechanism

$$X := f_X(\mathbf{X}_{\text{Pa}_X}, N_X) \xrightarrow{\text{do}(X=f_X^*(\mathbf{X}_{\text{Pa}_k^*}, N_X^*))} X := f_X^*(\mathbf{X}_{\text{Pa}_k^*}, N_X^*)$$

Example Linear SCM

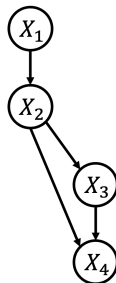
Given $N_i \sim \mathcal{N}(0, 1)$, define the linear SCM $\mathcal{C}$:

$$X_1 := \beta_{0,1} + N_1$$

$$X_2 := \beta_{0,2} + \beta_{1,2}X_1 + N_2$$

$$X_3 := \beta_{0,3} + \beta_{1,3}X_2 + N_3$$

$$X_4 := \beta_{0,4} + \beta_{1,4}X_2 + \beta_{2,4}X_3 + N_4$$



What happens to our linear SCM after intervening $do(X_3 = 1)$?

Example Linear SCM

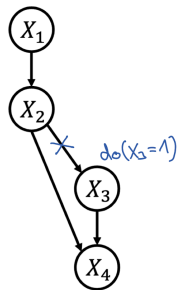
Given $N_i \sim \mathcal{N}(0, 1)$, the SCM $\mathcal{C}$ (after intervening on X_3)

$$X_1 := \beta_{0,1} + N_1$$

$$X_2 := \beta_{0,2} + \beta_{1,2}X_1 + N_2$$

$$X_3 := 1$$

$$X_4 := \beta_{0,4} + \beta_{1,4}X_2 + \beta_{2,4}X_3 + N_4$$



entails the distribution $P_{\mathbf{X}}^{\mathcal{C}; do(X_3=1)}$.

Note that:

- The intervention changed the structural assignment for X_3 , but left all other assignments **invariant**.
- All **edges from parent** nodes of 3 are deleted in the graph entailed by the new SCM.

Mathematical Definitions

Mathematical notation of manipulation (by Pearl):

- $do(X = x)$ represents a hypothetical **intervention** where X is set to the value x , uniformly over the entire population
- $p(y \mid do(X = x))$ is the **density** of Y after $do(X = x)$
- $E[Y \mid do(X = x)]$ is the **expectation** of Y after $do(X = x)$
- $P_X^{C; do(X=x)}$ is the **distribution** P_X^C **entailed by SCM** C after $do(X = x)$, etc.

It is important to **distinguish observation and interventions**:

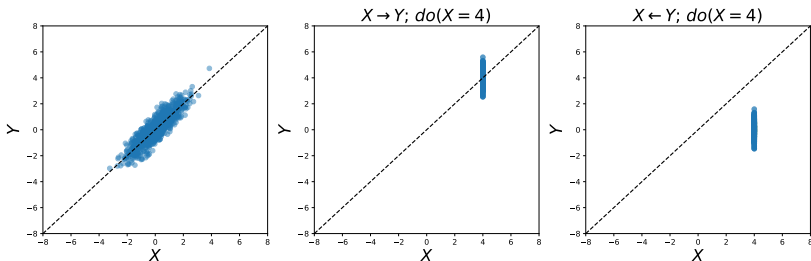
Conditioning on observing: $p(x \mid \text{see}(X = x)) = p(y \mid x)$

is generally **not equal** to

Intervening: $p(y \mid do(X = x))$, also written as $p^{do(X=x)}(y)$

Regression vs. Prediction of Interventions

- Given that we observe i.i.d. samples of $P(X, Y)$ (left).
- Our goal is to predict aspects of $p(y|x)$, e.g., $E[Y|X = x]$ (dashed).
- Regression is useful for prediction, but what if **we set X to 4**, i.e., $do(X = 4)$?



Observational data is generally **not useful** for drawing inferences about **interventional distributions**.

Intervening is not Observing

- Here, $p(y \mid do(X = x)) = p(y \mid do(X = x'))$ is the same for two different interventions x, x' on X .
- Since, X and Y are **correlated** in the **observed distribution**, however, $p(y \mid X = x) \neq p(y \mid X = x')$.

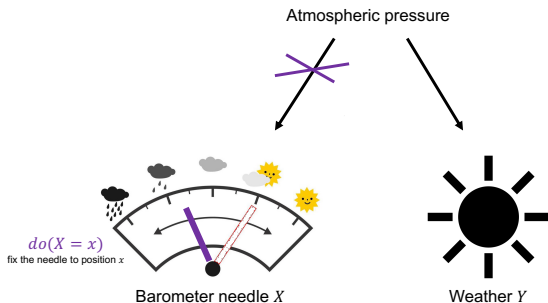


Figure 3: Credits Frederick Eberhardt

Causal Bayesian Networks

Recall: Bayesian Networks

Definition A **DAG model** or a **Bayesian network** is a combination $(\mathcal{G}, P)$, where distribution $P_{\mathbf{V}}$ with density p factorizes as

$$p(\mathbf{x}_{\mathbf{V}}) = \prod_{j \in \mathbf{V}} p(x_j | \mathbf{x}_{\text{Pa}(j)})$$

relative to the DAG $\mathcal{G} = (\mathbf{V}, \mathcal{E})$, i.e., $P_{\mathbf{V}}$ is **markov** relative to $\mathcal{G}$.

Causal Bayesian Networks (CBN)

Definition The pair $(\mathcal{G}, P)$ is a **causal DAG model** or a **causal Bayesian network** if for any $W \subset V$

$$p(x_V | do(X_W = x_w)) = \prod_{j \in V \setminus W} p(x_j | x_{Pa(j)}) [X_W = x_w],$$

where $[\cdot]$ is the indicator function. For continuous variables, we use the Dirac delta function.

A **causal Bayesian network** remains **well-defined under interventions**, and allows us to **compute interventional distributions**.

Properties of CBNs

An intervention on X_j only changes $p(x_j|x_{\text{Pa}(j)})$, but leaves the other conditional distributions unchanged. This is known as **invariance**.

Example:

- We know *altitude* influences *temperature* ($A \rightarrow T$).
- Thus, we can express $p(a, t) = p(t|a)p(a)$.
- $p(t|a)$ is a **physical law**, which we know is **invariant**.
- Changing the altitude does not change $p(t|a)$.

Post-Interventional Distributions

The pair $(\mathcal{G}, P)$ is a **causal Bayesian network** if for any $W \subset V$

$$\underbrace{p(x_V | do(X_W = x_w))}_{\substack{\text{post-interventional distribution} \\ \text{needed to define causal effects}}} = \prod_{j \in V \setminus W} \underbrace{p(x_j | x_{Pa(j)})}_{\substack{\text{conditional distribution that can} \\ \text{be computed from observational data}}} [X_W = x_w] .$$

Causal DAGs imply strong assumptions, allowing us to estimate *post-intervention distributions* from *observational data*.

How do we know the causal DAG?

Now: assume it is given, e.g., from background/domain knowledge

Later: consider learning causal DAGs (under some assumptions)

In any case, causal DAG provides a clear framework to **state causal assumptions** for analysis:

- Allows for an honest debate about such assumptions
- Can draw several possible causal DAGs, conduct analysis for each of them and perform a sensitivity analysis

Summary: Graph to CBN

- **DAG**: A directed graph without directed cycles.
- **Bayesian Network**: A combination $(\mathcal{G}, P)$, where $\mathcal{G}$ is a DAG and

$$p(x_{\mathbf{V}}) = \prod_{j \in \mathbf{V}} p(x_j | x_{\text{Pa}(j)}) .$$

- **Causal Bayesian Network**: A combination $(\mathcal{G}, P)$, where $\mathcal{G}$ is a DAG and for all $\mathbf{W} \subset \mathbf{V}$,

$$p(x_{\mathbf{V}} | do(X_{\mathbf{W}} = x_{\mathbf{w}})) = \prod_{j \in \mathbf{V} \setminus \mathbf{W}} p(x_j | x_{\text{Pa}(j)}) \Big|_{X_{\mathbf{W}} = x_{\mathbf{w}}} ,$$

The modified factorization is also known as:

- **"g-formula"** (Robins)
- **"manipulation formula"** (Spirtes, Glymour, Scheines)
- **"truncated factorization formula"** (Pearl)

SCMs and Causal DAG Models

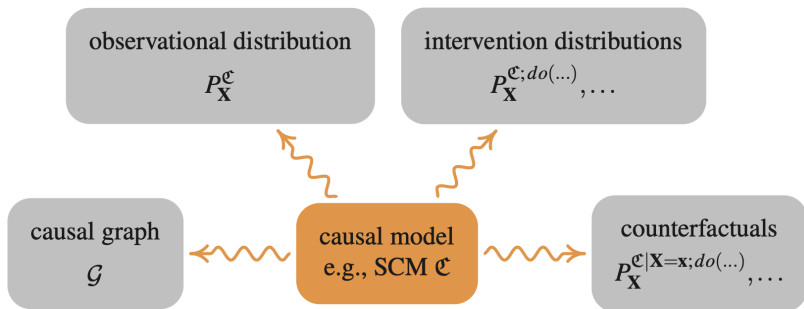


Figure 4: Figure 6.2 (ECI book): Causal models as SCMs do not only model an observational distribution P (Proposition 6.3) but also intervention distributions (Section 6.3) and counterfactuals (Section 6.4).

Causal vs. Statistical Notions

Statistical Parameter:

- Based on the joint **distribution**
- Makes no assumptions about *unobserved variables*
- Examples are $E[Y|X = x]$, $p(y = 1|x = 0)$, ρ_{XY} , etc.

Statistical Assumption:

- Constraint w.r.t. **distribution** of *observed variables*
- E.g., multivariate normal, P is *markov* w.r.t. a DAG, etc.

Causal Parameter:

- Defined in terms of an **SCM**
- Examples are the coefficients (e.g., β_{12}), the number parents of a node, the expected value of Y under $do(X = 4)$, etc.

Causal Assumptions:

- Constraint on the **SCM**, e.g.,
 - Linearity
 - Uncorrelatedness of N_i, N_j
 - That x_3 does not appear in $f_4(x_{Pa_4}, n_4)$

Total Average Causal Effect

Recall:

X has a **causal effect** on Y , if **manipulating** X changes Y .

We can now formally define:

The **total average causal effect** is defined as

$$ACE(x, x') = E[Y \mid do(X = x)] - E[Y \mid do(X = x')] .$$

As a special case, we define the **average treatment effect** of binary treatment variable T on a target Y as

$$ATE = E[Y \mid do(T = 1)] - E[Y \mid do(T = 0)] .$$

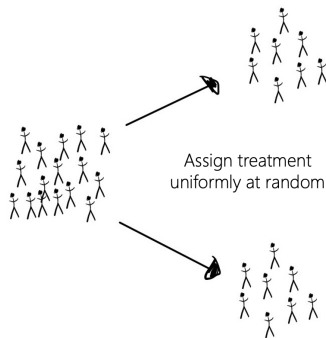
Interventions and RCTs

Randomized Controlled Trials

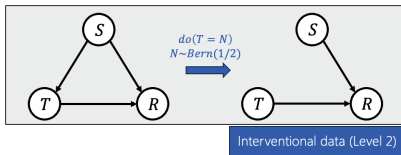
Gold standard for causal inference:

- A study that includes treated individuals and **controls**
- **Controlled experiment** and clear experimental design
- Treatment assignment is **randomized**
- The study is **double-blind**

Randomized Controlled Trials (Interventional Data)



RCT as an **intervention** on the data generative process



Such data allows for computing, e.g., **average treatment effects**:

$$\begin{aligned} ATE &= E[R \mid do(T = a)] - E[R \mid do(T = b)] \\ &\stackrel{\text{RCT}}{=} E[R \mid T = a] - E[R \mid T = b] \end{aligned}$$

What if we cannot do a RCT?

We will see in the next lecture how to use causal DAGs for computing post-interventional distributions!

- **Recommended reading** is *Elements of Causal Inference* Chapter 2.1, 3.1-2, 6.2-3, 6.5, and Chapter 1.3-5 in *Causality* (Pearl, 2009)
- This course *builds upon and extends* the material from “*Causality*” (WiSe 2022/23) that was kindly provided by **Stefan Harmeling**.
- This lecture, *builds upon and extends* the slides of the course “*Causality*” (2021) by Christina Heinze-Deml.

Thank you!